



Chapter 3

Cost

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Cost

(8 hours)

3.1 Cost classification

3.1.1 Total, average, fixed, variable, and marginal costs

3.1.2 Direct, indirect, and standard costs

3.1.3 Cash versus book cost, manufacturing and non-manufacturing cost

3.1.4 Sunk cost, opportunity cost, element of cost, life-cycle cost

3.2 Cost estimation and control

Total, average, fixed, variable, and marginal costs

Fixed Costs (FC): Expenses that remain constant regardless of production volume (e.g., rent).

Variable Costs (VC): Costs that change in direct proportion to production levels (e.g., raw materials).

Total Cost (TC): The sum of all costs: $TC = FC + VC$

Average Cost (AC): Total cost per unit of output:

$$AC = \frac{TC}{Q}$$

Marginal Cost (MC): The additional cost incurred by producing one more unit:

Total, average, fixed, variable, and marginal costs

Formula

$$MC = \frac{\Delta TC}{\Delta Q}$$

(and since FC doesn't change, also)

$$MC = \frac{\Delta VC}{\Delta Q}$$

If producing 10 m³ concrete costs 90,000
and 11 m³ costs 97,000

$$MC = 97,000 - 90,000 = 7,000$$

So, one extra m³ costs 7,000.

Total, average, fixed, variable, and marginal costs

Fixed Cost = 100,000

Variable Cost = 500Q

Total Cost:

$$TC = 100,000 + 500Q$$

Now find MC:

$$MC = \frac{d}{dQ}(100,000 + 500Q)$$

$$MC = 500$$

Calculus-Based Marginal Cost Formula:

$$MC = \frac{d(TC)}{dQ}$$

This means marginal cost is the derivative of total cost with respect to output.

Key point: Fixed costs do not affect marginal cost. Only variable costs change as output changes.

Total, average, fixed, variable, and marginal costs

Discrete Method (Using Change in Total Cost)

This method is used when cost data is provided for different levels of output.

Formula:

$$MC = \frac{TC_2 - TC_1}{Q_2 - Q_1}$$

- TC_2 = Total Cost at higher quantity
- TC_1 = Total Cost at lower quantity
- $Q_2 - Q_1$ = Change in output units

Total, average, fixed, variable, and marginal costs

Calculus-Based Method (Using Derivatives)

This method is used when the Total Cost (TC) function is given as an equation

Formula:

$$MC = \frac{d(TC)}{dQ}$$

Step-by-Step:

1. Express the total cost function as $TC=f(Q)$.
2. Differentiate the function with respect to Q .
3. Substitute the required output level into the derivative.

Total, average, fixed, variable, and marginal costs

A batching plant has total cost (Rs):

$$TC(Q) = 20000 + 120Q + 0.5Q^2$$

Find MC at $Q = 80$.

Step

- Marginal cost is derivative:

$$MC(Q) = \frac{dTC}{dQ} = 120 + Q$$

- At $Q = 80$:

$$MC(80) = 120 + 80 = 200$$

MC=Rs200 per unit at Q=80

Total, average, fixed, variable, and marginal costs

$$VC(Q) = 50Q + 0.8Q^2$$

Fixed cost $FC = 20000$. Find MC at $Q = 60$.

Step

$$TC = FC + VC \Rightarrow MC = \frac{dTC}{dQ} = \frac{dVC}{dQ}$$

$$MC(Q) = \frac{d}{dQ}(50Q + 0.8Q^2) = 50 + 1.6Q$$

At $Q = 60$:

$$MC(60) = 50 + 1.6(60) = 50 + 96 = 146$$

Total, average, fixed, variable, and marginal costs

For a precast yard:

$$TC(Q) = 40000 + 150Q + 0.6Q^2$$

Find **MC**, and comment if MC is increasing/decreasing with Q.

Step

$$MC(Q) = 150 + 1.2Q$$

To see trend, take derivative of MC:

$$\frac{dMC}{dQ} = 1.2 > 0$$

Answer: MC increases as Q increases (because slope is positive).

Direct, indirect, and standard costs

Direct cost is the cost directly associated with producing a product or completing a specific project activity.

- Directly measurable
- Varies with production level
- Easily assigned to a project

Indirect cost refers to expenses that support project activities but cannot be traced to one specific project or product.

- Shared among several projects
- Usually called overhead cost
- Difficult to allocate precisely

Standard cost is the expected or predetermined cost of producing a product or completing a project activity under normal conditions.

- Determined before production
- Used for budget planning
- Used to compare with actual cost

Formulas

- Profit = Total Revenue – Total Cost
- Total Revenue (TR) = Price × Quantity
- Break-even Quantity = Fixed Cost / (Selling Price – Variable Cost per unit)
- Break-even Revenue = Fixed Cost / Contribution Margin Ratio
- Contribution per Unit = Selling Price – Variable Cost
- Contribution Margin Ratio = (Selling Price – Variable Cost) / Selling Price

Formulas

- $\text{Operating Cost} = \text{Total Cost} - \text{Initial Investment}$
- $\text{Life Cycle Cost (LCC)} = \text{Initial Cost} + \text{Operating Cost} + \text{Maintenance Cost} + \text{Disposal Cost}$
- $\text{Cost Variance} = \text{Actual Cost} - \text{Estimated Cost}$
- $\text{Cost Performance Index (CPI)} = \text{Earned Value} / \text{Actual Cost}$
- $\text{Estimated Cost at Completion (EAC)} = \text{Budget} / \text{CPI}$

Direct, indirect, and standard costs

Total Project Cost

$$\text{Total Cost} = \text{Direct Cost} + \text{Indirect Cost}$$

Standard Cost Variance

$$\text{Cost Variance} = \text{Actual Cost} - \text{Standard Cost}$$

If variance **positive** → cost overrun

If variance **negative** → cost saving

Direct, indirect, and standard costs

Total Project Cost

$$\text{Total Cost} = \text{Direct Cost} + \text{Indirect Cost}$$

Standard Cost Variance

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If variance **positive** → cost overrun

If variance **negative** → cost saving

Direct, indirect, and standard costs

➤ A factory has the following standard data:

- Standard number of workers = 100
- Standard wage rate = Rs 1.60 per hour
- Standard output = 40 units per hour
- Hours paid per week = 80 hours

During a week:

- Actual output = 2000 units
- Idle time = 20%
- Actual wage rate = Rs 1.80 per hour

Calculate labour cost variance.

Direct, indirect, and standard costs

Solution

Standard production per hour

$$40 \text{ units/hour}$$

Total standard hours required

$$\frac{2000}{40} = 50 \text{ hours}$$

Actual wage cost

$$100 \times 80 \times 1.80 = 14,400$$

Direct, indirect, and standard costs

Standard wage cost

$$100 \times 80 \times 1.60 = 12,800$$

Labour variance

$$14,400 - 12,800 = 1,600$$

Answer

Labour cost variance = Rs 1,600 unfavorable

Direct, indirect, and standard costs

➤ A factory uses two materials A and B.

Material	Quantity	Standard Price
A	48 kg	Rs 10/kg
B	112 kg	Rs 2/kg

Find standard material cost for the production batch.

Direct, indirect, and standard costs

Solution

Material A cost

$$48 \times 10 = 480$$

Material B cost

$$112 \times 2 = 224$$

Total standard material cost

$$480 + 224 = 704$$

Answer

Standard material cost = **Rs 704**

Direct, indirect, and standard costs

- Factory overheads are applied at 60% of direct wages. If direct wages = Rs 50,000, calculate: Factory overhead and Total factory cost.

Solution

Factory overhead

$$0.60 \times 50,000 = 30,000$$

Total factory cost

$$50,000 + 30,000 = 80,000$$

Answer

Factory overhead = Rs 30,000

Factory cost = Rs 80,000

Direct, indirect, and standard costs

➤ Indirect costs of a construction firm are:

- Administration = Rs 30,000

- Insurance = Rs 10,000

- Office rent = Rs 20,000

These costs are allocated to 5 projects. Find indirect cost per project.

➤ Standard cost of producing one product = Rs 120

Actual production = 500 units

Actual total cost = Rs 65,000

Find:

Total standard cost

Cost variance

Cash versus book cost, manufacturing and non-manufacturing cost

Cash Cost	Book Cost
Cash cost refers to the actual cash outflow paid by a firm for expenses during production or operation.	Book cost refers to accounting expenses recorded in financial statements but not involving actual cash payment at that time.
Involves real cash payment.	No actual cash payment occurs at the time of recording.
Recorded as cash expenditure in financial accounts.	Recorded as non-cash expense for accounting purposes.
Directly reduces the firm's cash balance.	Does not affect current cash flow.
Used to analyze actual financial liquidity and operational spending.	Used to represent asset usage or allocation of cost over time.
Occurs when the payment is made.	Occurs periodically through accounting entries.
Important for short-term financial decisions and cash budgeting.	Important for financial reporting, tax calculation, and asset valuation.
Wages paid to workers, raw material purchase, electricity bills, fuel cost.	Depreciation of machines, amortization of assets, depletion of natural resources.
Decreases profit and cash simultaneously.	Decreases accounting profit only but not cash.
Used to calculate actual operating costs of projects.	Used to calculate depreciation and tax benefits in project evaluation.

Cash versus book cost, manufacturing and non-manufacturing cost

Manufacturing Cost	Non-Manufacturing Cost
Manufacturing costs are the costs directly related to the production of goods in a factory.	Non-manufacturing costs are the costs not related to the production process, but associated with selling, administration, and business operations.
Used to produce or manufacture a product.	Used to manage, sell, and distribute the product.
Occurs inside the factory or production facility.	Occurs outside the production process.
Directly related to manufacturing activities.	Indirectly related or not related to production.
Considered product cost.	Considered period cost.
Included in cost of goods manufactured (COGM) and inventory valuation.	Recorded as operating expenses in the income statement.
Direct materials, direct labor, manufacturing overhead.	Selling expenses and administrative expenses.
Helps determine cost of production and product pricing.	Helps determine operating expenses and profit analysis.
Important for production efficiency and cost control.	Important for business management and marketing control.
Factory floor, production unit, assembly line.	Office, marketing department, sales department.

Sunk cost, opportunity cost, element of cost, life-cycle cost

Cost Concept	Definition	Main Idea / Meaning	Key Characteristics
Sunk Cost	A cost that has already been incurred in the past and cannot be recovered.	It should not affect future decision making because the money is already spent.	Past cost ,Irrecoverable , Ignored in economic analysis
Opportunity Cost	The value of the best alternative that is sacrificed when a decision is made.	It represents the benefit you lose by choosing one option over another.	Related to alternatives, Important for decision making , Not always recorded in accounting books
Element of Cost	The basic components that make up the total cost of producing a product or service.	Helps in cost estimation and cost control in engineering projects.	Usually classified into: Material cost, Labor cost Expenses / overhead
Life-Cycle Cost (LCC)	The total cost of an asset over its entire life, from purchase to disposal.	Considers all costs during the life of a system, not just the initial cost.	Includes: Initial cost Operating cost, Maintenance cost, Replacement cost, Disposal cost

Cost estimation and control

Cost estimation is the process of forecasting the total cost required to complete a project before its execution.

- Budget preparation
- Tender submission
- Project feasibility analysis
- Resource planning
- Financial approval

Cost control is the process of monitoring, comparing, and regulating actual project costs to ensure they remain within the approved budget.

- Prevent cost overrun
- Improve efficiency
- Minimize wastage
- Maintain profitability

Cost estimation and control

Aspect	Cost Estimation	Cost Control
Definition	Cost estimation is the process of predicting the total cost required to complete a project or produce a product before the work begins.	Cost control is the process of monitoring and managing project expenses to ensure the project is completed within the planned budget.
Purpose	To determine the expected cost of resources such as materials, labour, equipment, and overhead.	To prevent cost overruns and keep project spending within the budget.
Stage in Project	Used before the project begins, mainly during planning and design stages.	Used during project execution and monitoring stages.
Concern	Predicting and planning costs.	Controlling and managing actual costs.
Objective	To prepare budget, feasibility analysis, and financial planning.	To compare actual cost with estimated cost and take corrective action if needed.

Cost estimation and control

Data Used	Historical project data, engineering drawings, material prices, labour rates, and equipment costs.	Actual project expenses, invoices, labour records, and financial reports.
Main Components Considered	Material cost, labour cost, equipment cost, overhead cost, contingency cost.	Monitoring expenses, cost comparison, cost reporting, corrective action.
Techniques / Methods	Analogous estimation, parametric estimation, bottom-up estimation, top-down estimation.	Budget monitoring, variance analysis, performance measurement, corrective actions.
Importance	Helps in project budgeting, bid preparation, and economic feasibility analysis.	Helps in maintaining financial discipline and improving project efficiency.
Decision Making Role	Helps choose the most economical project alternative.	Helps ensure the selected project remains financially viable during execution.

Cost estimation and control

Estimation involves predicting future costs for budgeting and project management.

Core Estimation Formulas:

- **High-Low Method (Splitting Mixed Costs):**
 - **Variable Cost/Unit** = $(\text{High Cost} - \text{Low Cost}) / (\text{High Units} - \text{Low Units})$.
 - **Fixed Cost Component** = $\text{Total Cost} - (\text{Variable Cost/Unit} \times \text{Units})$.

Cost estimation and control

Key Mathematical Models:

1. **The High-Low Method:** Used to separate mixed costs into fixed and variable components.

$$1. \text{ Variable Cost per unit (v)} = \frac{\text{Highest Cost} - \text{Lowest Cost}}{\text{Highest Activity} - \text{Lowest Activity}}$$

$$2. \text{ Fixed Cost (f)} = \text{Total Cost} - (v \times \text{Total Units})$$

2. **Linear Regression (Least Squares):** In 2026, software is typically used to solve the equation $Y = a + bX$, where Y is total cost, a is fixed cost, b is variable rate, and X is the activity level.

Cost estimation and control

Learning Curve Estimation: Estimates how labor costs decrease as workers gain experience.

1. Formula: $Y = aX^b$ (where Y is the cumulative average time per unit).

Control Variances: Used to manage costs by comparing actuals to standards.

1. **Cost Variance**

$$= (\text{Standard Price} \times \text{Standard Quantity}) - (\text{Actual Price} \times \text{Actual Quantity})$$

Questions

➤ A workshop has the following maintenance cost data:

- Highest machine hours = 800 hours, cost = Rs 18,000
- Lowest machine hours = 500 hours, cost = Rs 12,000

Find:

- Variable cost per hour
- Fixed cost
- Cost equation

Questions

Step 1: Find variable cost per hour

$$b = \frac{18000 - 12000}{800 - 500}$$

$$b = \frac{6000}{300} = 20$$

- Variable cost per hour = Rs 20

Step 2: Find fixed cost

Using highest activity data:

$$a = 18000 - (20 \times 800)$$

$$a = 18000 - 16000 = 2000$$

Questions

- Fixed cost = Rs 2,000

Step 3: Cost equation

$$Y = 2000 + 20X$$

- Variable cost per hour = Rs 20
- Fixed cost = Rs 2,000
- Cost equation = $Y = 2000 + 20X$

Questions

A plant produced:

- 3,000 units costing \$59,000
- 1,250 units costing \$38,000

Find

1. Variable cost per unit
2. Fixed cost
3. Cost function

Questions

Step 1: Variable cost

$$\frac{59000 - 38000}{3000 - 1250}$$
$$= \frac{21000}{1750} = 12$$

Variable cost per unit = \$12

Step 2: Fixed cost

$$59000 - (12 \times 3000)$$
$$59000 - 36000 = 23000$$

Step 3: Cost function

$$Y = 23000 + 12X$$

Questions

1. A company records maintenance costs as follows:
Highest activity = 900 machine hours, cost = Rs 27,000
Lowest activity = 500 machine hours, cost = Rs 17,000
Determine the variable cost per machine hour and fixed cost using the high-low method.
2. A manufacturing firm produced 1,800 units at a cost of Rs 72,000 and 1,200 units at a cost of Rs 54,000. Estimate the cost function using the high-low method.
3. A workshop reports electricity costs as:
Highest usage = 2,500 kWh, cost = Rs 18,500
Lowest usage = 1,600 kWh, cost = Rs 13,400
Find the variable cost per unit and fixed cost.

Questions

4. A firm has the cost function $C(Q) = 1500 + 6Q + 0.2Q^2$. Find the marginal cost when $Q = 20$ units.
5. The total cost function of a company is $C(Q) = 1000 + 5Q + Q^2$. Find the marginal cost when output is 10 units.
6. The total cost of producing 50 units is Rs 3,500 and the cost of producing 60 units is Rs 4,100. Calculate the marginal cost per unit between these levels of production.
7. A manufacturing firm has the cost function $C(Q) = 250 + 12Q + 0.1Q^2$. Find the marginal cost function and marginal cost when $Q = 30$.

Thank you for your attention!